GAJA SAVIYA

Open data and AI simulation to map elephant risks and route rural Sri Lankans safely.



Maps elephant-human spatial overlap zones.

Filters sparse pings into paths

Calculates risk-adjusted path travel options.

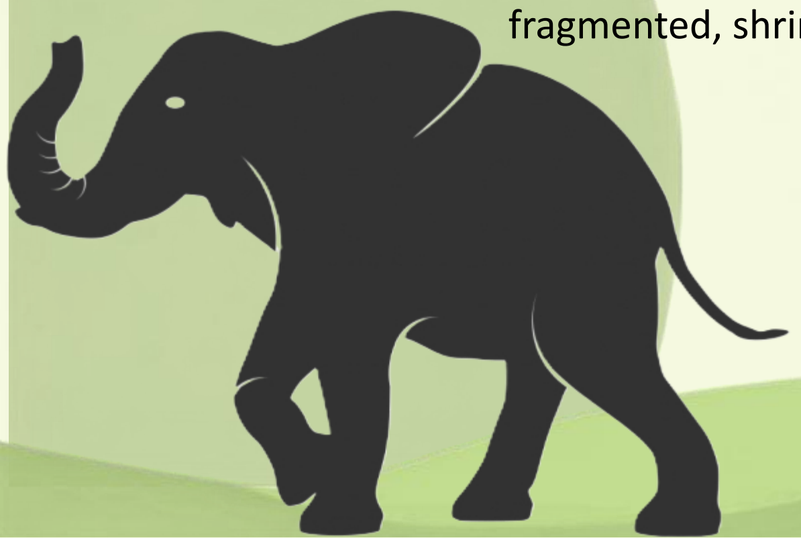
Displays alerts and interactive routes.

Open-data AI framework that simulates elephant movements to deliver real-time collision alerts and safe-routing for rural Sri Lanka.

GAJA SAVIYA

The Problem

- **Tragic Loss of Life:** Sudden encounters lead to severe casualties for both humans and elephants.
- **Devastating Economic Loss:** Crop destruction wipes out farmers' livelihoods, while property damage drains local resources.
- **Constant Fear & Habitat Loss:** Communities live under restricted mobility, while elephants face fragmented, shrinking habitats.

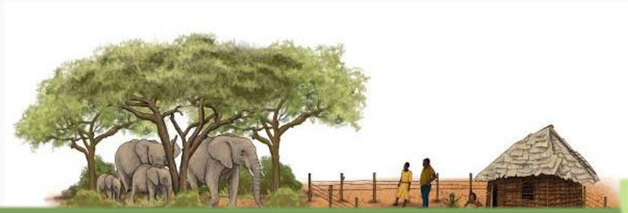


Our Solution — GAJA SAVIYA

Open data and AI simulation to map elephant risks and route rural Sri Lankans safely.

1. **Open Geospatial Data**
   Collect public habitat, population, and road map layers
2. **Overlap Risk Mapping**
   Calculate conflict zones using human-habitat spatial AI
3. **Sparse Motion Tracking**
   Estimate hidden elephant paths and location uncertainty
4. **Risk-Aware Safe Routing**
   Generate alternative paths avoiding high-danger grid zones
5. **Live Map Alerts**
   Display real-time danger regions, paths, and village warnings

Research Gap



EXISTING APPROACHES	RESEARCH GAPS	PERICARE'S NOVEL APPROACH
Costly field hardware (camera traps, sensors)	High cost, hard to maintain, and difficult for university teams to deploy	Software-only framework using open datasets and simulation
Official restricted collar data & incident coordinates	Restricted and unavailable for student research	Habitat-constrained movement simulator with transferred open statistics
Static hotspot maps	Gap between static maps and actionable alerts/routes	Integrated infant activity & development tracking
Sparse collar GPS updates	Unknown animal location and position between pings	Dynamic risk surface combined with safe-routing and alerts
Standard shortest-path routing	Directs travel through high-risk elephant-human overlap cells	Extended Kalman Filter tracker estimating position and uncertainty
Post-incident warnings	Reactive warning provided only after an elephant enters a village	Dynamic road-network planner prioritizing safer routes over shorter ones

KEY GAP

Existing research and systems either rely on inaccessible hardware and restricted collar datasets or stop at static hotspot maps, lacking an integrated, software-only framework that combines open-data spatial risk mapping, sparse location tracking, dynamic safe routing, and live map alerts for rural Sri Lanka..

Main Objective

To design, implement, and evaluate a software prototype that estimates human–elephant conflict risk from open data, tracks simulated sparse wildlife locations, and generates safer village-to-town routes and geofence alerts for a Sri Lankan study area.

01 MAP RISK

02 SIMULATE MOVEMENT

03 TRACK UNCERTAINTY

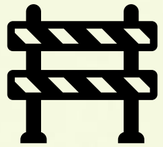


04 PLAN SAFE ROUTES

05 INTEGRATE DEMO INTERFACE

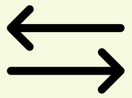
Current Challenges During Development

01



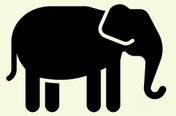
RESTRICTED & BIAS DATA

02



MOVEMENT TRANSFER VALIDATION

03

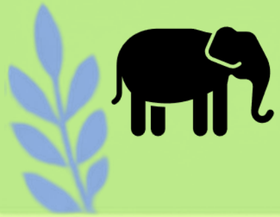


UNCERTAINTY CALIBRATION

04



SYSTEM INTEGRATION



Overall Challenge:
Combining spatial risk, sparse movement tracking, and dynamic routing into an open-data software prototype while ensuring transparency, realistic simulation, and actionable community safe paths.

COMMERCIALIZATION PLAN

Providing an open-data, software-only framework for early warning and safe-routing for human-elephant conflict in rural Sri Lanka.



TARGET MARKET:

- Rural communities in HEC corridors;
- District authorities.

VALUE PROPOSITION:

- Reduced human death;
- Actionable early warnings;
- Scalable software method.

REVENUE MODEL:

- Software-as-a-Service for authorities;
- Data licensing for NGOs;
- Research grants.

COST ESTIMATION:

- Cloud infrastructure;
- Algorithm maintenance;
- Data acquisition (open).

PANEL COMMENTS & IMPROVEMENTS

Analysis of feedback from Panel IT-P10 and the subsequent system-wide improvements made for Gaja-Saviya.

	PANEL COMMENT	IMPROVEMENT MADE
1	**Research Focus and Technical Understanding** Demonstration and Q&A reflected insufficient understanding of the research direction, objectives, and proposed technical implementation.	**Deepened Research Scope & Clear Objectives** • Refined project objectives to create a software-only HEC early-warning framework. • Conducted deep-dive research into software methods for spatial risk and sparse tracking.
2	**Technical Implementation and Depth** Technical implementation expected for this project was not clearly demonstrated or justified during the panel review.	**Advanced Software Architecture Implemented** • Implemented an Extended Kalman Filter (EKF) tracker for sparse location estimation. • Built a dynamic, risk-weighted road network planner using OpenStreetMap data.
3	**Project Progress and Milestones** Lack of visible experimental progress and an overall lack of clarity on technical milestones for the project.	**Visible System Demonstration and Validation** • Integrated Component 1 (Risk Map) and Component 2 (Tracking) into a live map demo. • Validated risk surface against published Sri Lankan conflict hotspots.
4	**Student Group 4 Cohesion (Individual and Group-level Comments)** Feedback for IT22091598, IT22249098, IT22145734 was critical; feedback for IT22544636 was positive for experimental progress.	**Integrated Team Development** • Implemented cross-functional team tasks to elevate whole group understanding. • Shared individual experimental results (e.g., student 4) to boost overall project quality.

THANK YOU

ARE THERE ANY QUESTIONS?

